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April 30th 2026

Data Project: Mental Health Survey Results

Covid was an interesting time period for mental health in the United States. In order to explore this topic, I chose a data set from the U.S. Census Bureau that contained survey results for whether or not American households were utilizing mental healthcare - either pharmaceutical or counseling. Exploring this data set to serve my personal curiosity relating to how different groups of people were reacting during this time.

This data was collected by the Household Pulse Survey, which was a collaboration between the U.S. Census Bureau and five federal agencies in order to produce data on the social and economic impacts of Covid-19 on American Households. Data was collected via an internet questionnaire, which was sent out via email and text. Data estimates were adjusted for non-response and to match the Census Bureau's estimates in accordance with the NCHS Data Presentation Standards for Proportions. With this analysis, I hope to parse this dataset into meaningful results that show insight into how American households across the country dealt with Covid-19.

After some initial analysis, I hoped to answer the following questions:

1: How did the use of therapy and medication change over time

2: How did different states cope with social and economic stressors of Covid-19.

Methods

My chosen dataset was obtained via data.gov("Mental Health Care in the Last 4 Weeks"). This dataset represented portions of survey results collected 2020-2022 that pertained to how different populations responded to mental health indicators. Each row represents a survey result, which had a grouping, and the estimated percentage of people in said population that participated in one of the four indicators.

Mental Health Indicators - Last Four Weeks	Survey Groupings (By)
• Received Counseling or Therapy • Needed Counseling or Therapy But Did Not Get It • Took Prescription Medication for Mental Health • Took Prescription Medication for Mental Health And/Or Received Counseling or Therapy	Sex, Age, National Estimate, Presence of Symptoms of Anxiety/Depression, Race/Hispanic ethnicity, Education, State, Sexual orientation, Disability status, Gender identity

Figure 1.1

Initially, I cleaned the dataset, removing duplicate columns (multiple columns describing survey time period and confidence intervals). I then refactored the time columns to be a start date and a duration, as opposed to a start and end date. With this cleaned data, I aggregated and plotted how the indicators changed for all groupings to get a general sense of how survey participants' responses changed over time.

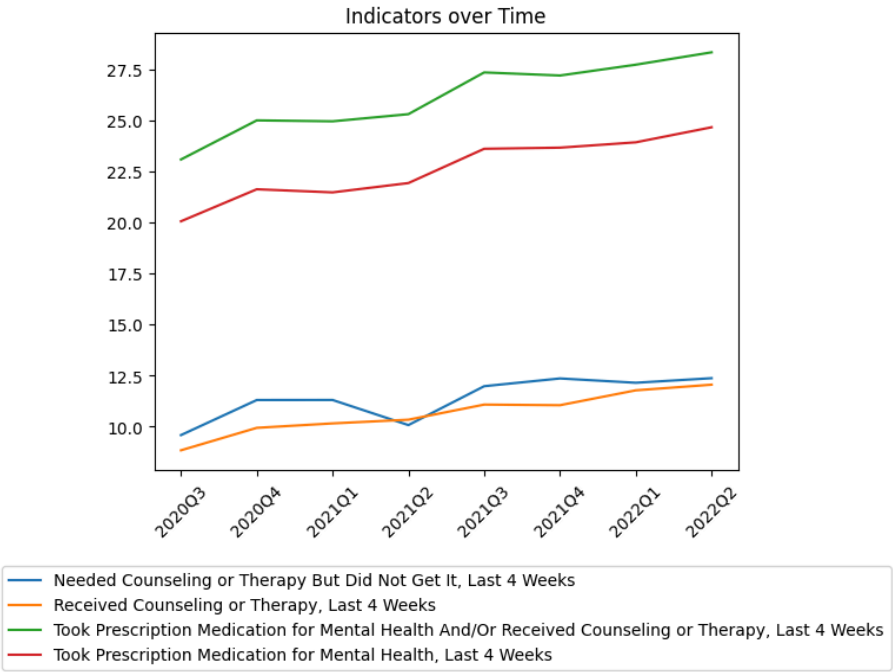


Figure 1.2

I then inspected the row counts for each grouping, in order to see if there was a singular grouping set that took up most of the overall dataset.

Grouping	Row Count
By Sex	304
By Age	1064
National Estimate	152
By Presence of Symptoms of Anxiety/Depression	304
By Race/Hispanic ethnicity	760
By Education	608
By State	6732
By Sexual orientation	156
By Disability status	168
By Gender identity	156

Figure 1.3

From figure 1.3, I surmised that the most promising grouping for analysis was the by state grouping, which had a subgroup column representing where the survey was collected (all U.S. states and District of Columbia).

In order to analyze the grouped by states data, I first pivoted the frame such that each row represented a state, and then their aggregated survey results for each of the four indicators. With this, I scaled the data and used sklearn's PCA class to find the principal components. From these results, I created a scree plot to determine how many principal components to select.

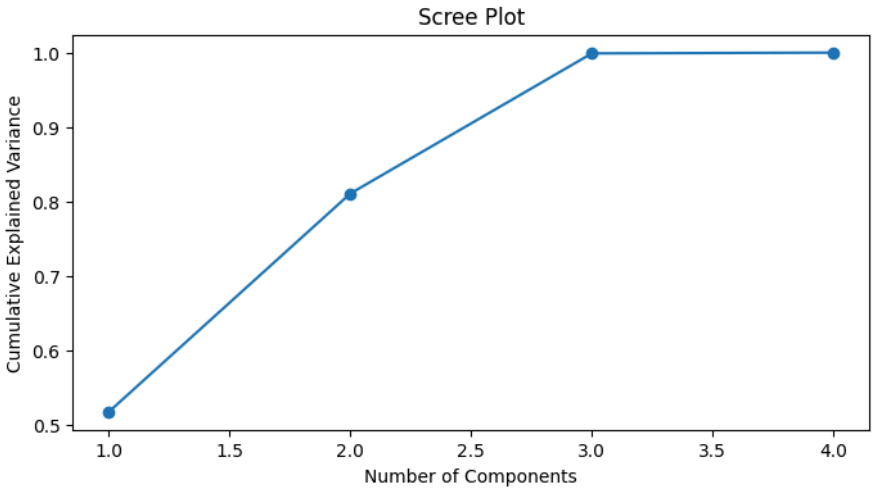


Figure 2.1

I selected two principal components, as just two described over 80% of the matrix variance.

Principle Component Analysis Results

Indicator	PC1	PC2
Took Prescription Medication for Mental Health	0.681971	-0.099613
Took Prescription Medication and Received Counseling	0.616967	-0.423803
Needed Counseling or Therapy But Did Not Get It	0.324608	0.508897
Received Counseling or Therapy	0.221126	0.742625

Figure 2.2

PCA1 explains all of the behavior regarding taking medication for mental health in the past 4 weeks, therefore the PCA1 axis of the PCA space represents latent sentiment towards medication. PCA2 explains all of the behavior regarding received or wanting therapy, therefore the PCA2 axis of the PCA space represents latent sentiment towards counseling - both receiving it, and needing it and not receiving it.

I next plotted each state in the PCA space:

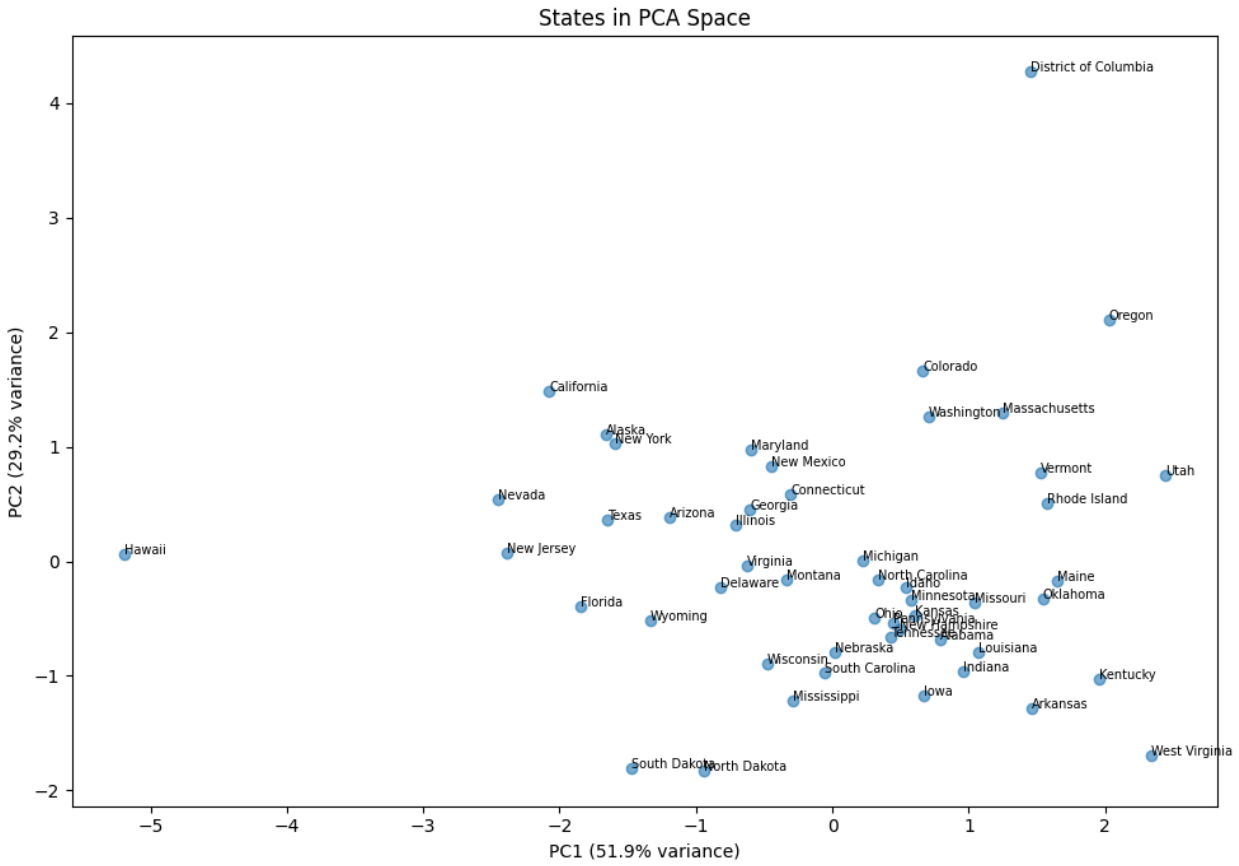


Figure 2.3

K-Means

Elbow method:

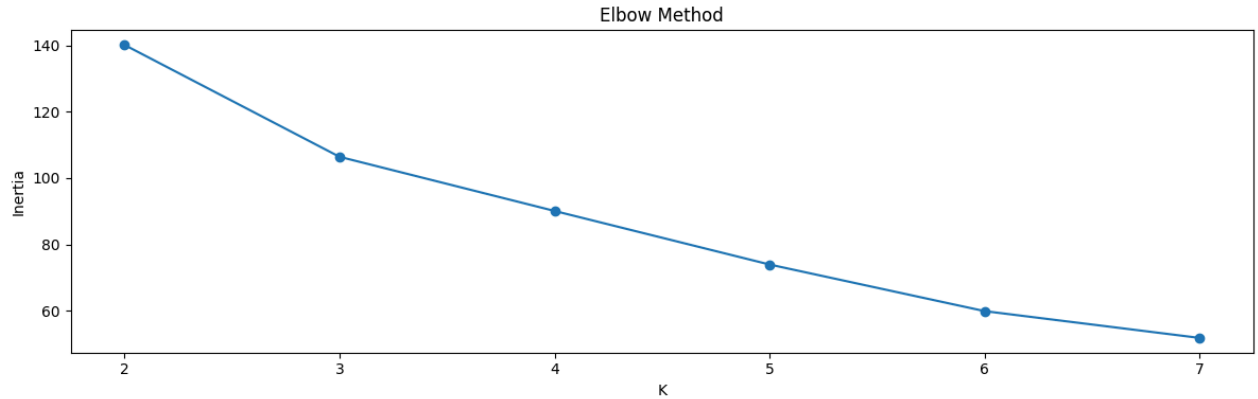
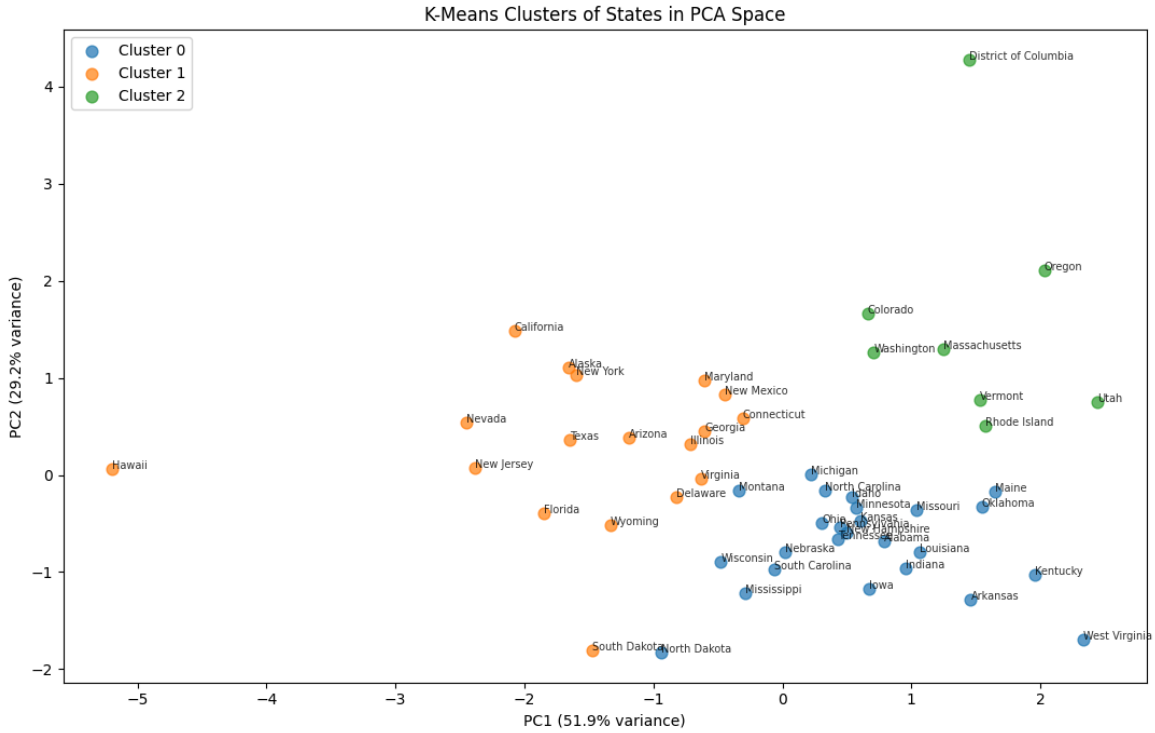


Figure 3.1

From inspection of figure 3.1, we can see a change in slope at 3 clusters. This is not the most explicit result of the elbow method, as increasing past 3 clusters only results in slightly diminishing returns on inertia.

Analysis of cluster



Results of K-means clustering

Above, we can see that in our PCA space, with three clusters we have some clear grouping for different states' latent sentiment towards medication and therapy for mental health between 2020 and 2022. Outliers: District of Colombia is extremely high in both principal components, indicating that therapy is accepted and medication is normalized. Hawaii in contrast did not flag high in either dimension.

Group	Contents
Group 0	'Alabama', 'Arkansas', 'Idaho', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebraska', 'New Hampshire', 'North Carolina', 'North Dakota', 'Ohio', 'Oklahoma', 'Pennsylvania', 'South Carolina', 'Tennessee', 'West Virginia', 'Wisconsin'
Group 1	'Alaska', 'Arizona', 'California', 'Connecticut', 'Delaware', 'Florida', 'Georgia', 'Hawaii', 'Illinois', 'Maryland', 'Nevada', 'New Jersey', 'New Mexico', 'New York', 'South Dakota', 'Texas', 'Virginia', 'Wyoming'
Group 2	'Alaska', 'Arizona', 'California', 'Connecticut', 'Delaware', 'Florida', 'Georgia', 'Hawaii', 'Illinois', 'Maryland', 'Nevada', 'New Jersey', 'New Mexico', 'New York', 'South Dakota', 'Texas', 'Virginia', 'Wyoming'

Secondary analysis: Cross referencing vs. mean cluster GDP per capita

Cluster	Mean GDP per capita
0	76288
1	92449.17
2	115151.12

Figure 4.1

With these groupings, I was curious to cross reference against other information to see if there were any likely explanations for why these groups of states responded the way they did. Doing some cursory analysis by taking state GDP per capita data into account, we can see that our groups have very

different levels of wealth. Group 0, which had high sentiment towards medication for mental health, but low sentiment towards therapy, was the poorest of the three groups, with a GDP per capita just north of $75,000. The wealthiest of the three groups, group 2 - with a GDP per capita higher around $115,000, had high sentiment for both therapy and pharmaceuticals for mental health treatment. In between the two at $92,000, group 1 had lower sentiment towards both treatment avenues (Wikipedia Contributors).

Conclusions

From this dataset, the following results became clear:

One: Through the time period of 2020-2022, survey responses indicating treatment of mental health conditions increased - and that for the entire time period, aggregated across all groups, responders leaned more towards pharmaceuticals than towards counseling to treat mental health conditions.

Two: States responded differently to mental health treatment surveys. Some states had higher latent sentiment towards medication, some had higher sentiment towards therapy. Interestingly, no states had high sentiment towards therapy, but low sentiment towards medication.

Going further, pull from more time relevant data for GDP - this was cursory data from wikipedia to get a general heuristic on how wealthy each state cluster was. Look at the timeline to see how the overall indicator fluctuates in relation to the events of Covid-19 pandemic - i.e. reopenings/shutdowns. Additionally, because of the nature of survey results, this analysis is dependent on how the Census Bureau accounted for non-responder bias. The state clustering analysis can only speak to the aggregation of the states generally, and cannot speak to individuals or individual groups in said states. Further work on this dataset could include analyzing and comparing the different groups, but this would be difficult as it is not an apples to apples comparison (By age vs History of mental health diagnosis).

Works Cited

"Mental Health Care in the Last 4 Weeks." *Data.gov*, Centers for Disease Control and Prevention, 18 May 2022, catalog.data.gov/dataset/mental-health-care-in-the-last-4-weeks.

Wikipedia Contributors. "List of U.S. States and Territories by GDP." *Wikipedia*, Wikimedia Foundation, 30 Jan. 2019, en.wikipedia.org/wiki/List_of_U.S._states_and_territories_by_GDP.